TEXORA · LECTURE SHEET

Process Control

Process control is the discipline of keeping every spinning process on its target value with minimum variation: measure, compare with the standard, act on the cause. This opening sheet sets out the statistical tools the rest of the course uses — mean, standard deviation, CV%, control charts and the index of irregularity — each with spinning data.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 1 · SPECIALIZATION — YARN MANUFACTURING

1. What process control means in a spinning mill

Every product of spinning — lap, sliver, roving, yarn — has target values (weight per unit length, count, strength, twist) and every real process wanders around its target. **Process control** is the organised work of (a) defining the target and its tolerance, (b) measuring the product at a planned frequency, (c) judging whether the process is behaving normally, and (d) correcting the *process* — settings, machine condition, material, climate — rather than sorting the product afterwards. It differs from inspection in timing and philosophy: inspection finds bad product; process control prevents it.

Variation has two kinds of cause. **Random (common) causes** are the small, always-present fluctuations of fibre properties, machine behaviour and measurement itself; together they set the process's natural spread. **Assignable (special) causes** — a worn roller, a wrong change wheel, a new mixing laid down badly — shift the mean or inflate the spread suddenly. Control charts exist to tell the two apart, so that the mill neither chases random noise nor misses a real fault.

2. Describing variation: mean, SD and CV%

$$\text{Mean } \bar{x} = \Sigma x / n \quad \cdot \quad \text{SD} = \sqrt{[\ \Sigma(x - \bar{x})^2 / (n - 1)\ ]} \quad \cdot \quad \text{CV\%} = \text{SD} / \bar{x} \times 100$$

WORKED EXAMPLE — A SLIVER WEIGHT TEST

Ten consecutive 1-metre cut lengths of drawframe sliver weigh (g/m): 3.90, 3.95, 3.88, 3.94, 3.90, 3.92, 3.89, 3.93, 3.91, 3.88.

Mean = 39.10 / 10 = **3.910 g/m**.

Standard deviation (from the squared deviations) = **0.0245 g/m**.

CV = 0.0245 / 3.910 × 100 = **0.63%**.

Against a nominal sliver of 3.90 g/m the mean is 0.26% heavy — well inside a typical ±1% action

band — and the CV of 0.63% shows the short-term weight variation is small. Both numbers are needed: the mean locates the process, the CV describes its steadiness.

3. The control chart

A control chart plots measurements in time order against a centre line (the process mean) and control limits at ±3 SD. Points inside the limits with no pattern indicate random variation only; a point outside, or a run of points drifting to one side, signals an assignable cause.

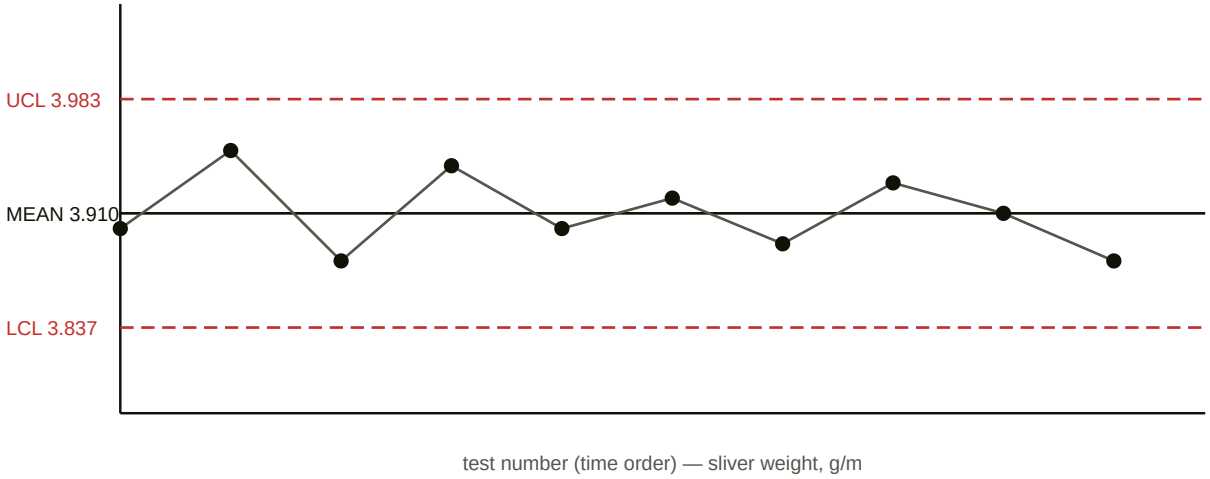


Figure 1 — Control chart of the worked-example sliver test. All ten points lie inside the ±3 SD limits with no run or trend: the process is in statistical control, and its 0.63% CV is the random-cause level.

4. Evenness: U%, CV% and the index of irregularity

Mass variation measured over very short lengths (the Uster evenness test) is quoted as U% or CV_m. There is a theoretical floor: even a perfectly random arrangement of fibres has irregularity, the **limiting irregularity**, set by fibre number and fineness in the cross-section. The ratio measures how much of the observed unevenness the process itself is adding:

$$\text{Index of irregularity} = \text{actual CV} / \text{limiting CV}$$

WORKED EXAMPLE — INDEX OF IRREGULARITY

A yarn shows CV_m = 5.8% (short-term, capacitance test) and the limiting CV for its fibre cross-section is 4.6%.

Index = 5.8 / 4.6 = **1.26**.

An index of 1.00 would mean the process adds nothing beyond random fibre arrangement — unattainable in practice. Values around 1.2–1.4 are typical of a healthy process; a rising index

points to drafting faults, roller problems or poor fibre control long before the yarn fails its specification.

5. The control loop in practice

STEP	ACTION	SPINNING EXAMPLE
Standard	Fix target and tolerance from the spin plan	Sliver 3.90 g/m ± 1%
Measure	Planned sampling, calibrated instruments, standard atmosphere	Cut-and-weigh test each shift
Judge	Chart the result; separate random from assignable variation	Point beyond UCL after a roller change
Act	Correct the process cause, not the individual reading	Re-set roller, verify with a fresh test
Record	Keep charts and actions for audit and trend analysis	Department control file

The topics that follow apply this same loop department by department — mixing, blow room, carding, draw frame, comber, simplex, ring frame, winding and rotor — and finish with the two mill-level controls: yarn realisation (is material becoming yarn?) and the machinery audit (are the machines fit to hold the process?).

Glossary

Process control — keeping a process on target by measurement, judgement and correction of causes. **Assignable cause** — a specific, findable source of abnormal variation. **Control limits** — the ±3 SD boundaries on a control chart. **CV%** — standard deviation as a percentage of the mean. **Limiting irregularity** — the unevenness of an ideal random fibre arrangement. **Index of irregularity** — actual CV divided by limiting CV. **U%** — mean deviation of mass, the older evenness measure, roughly CV% × 0.8 for normal distributions.

Key points

• Process control prevents faults by correcting causes; inspection only finds faulty product.
• Mean locates the process, CV% measures its spread (example: 3.910 g/m, CV 0.63%).
• Control limits at mean ± 3 SD separate random variation from assignable faults.
• Index of irregularity = actual/limiting CV (example 1.26); a rising index is an early warning.
• Every department in this course runs the same loop: standard → measure → judge → act → record.

Practice

1. Five roving weight tests give 0.488, 0.495, 0.491, 0.497, 0.489 g/m. Calculate the mean and CV% (use SD = 0.0039 g/m) and the ±3 SD control limits.
2. A process has mean 20.0 and SD 0.3. A test reads 21.1. Is an assignable cause indicated? Show the limit calculation.
3. Explain why reacting to every individual reading inside the control limits makes variation worse, not better.

Answer key

1. Mean = (0.488+0.495+0.491+0.497+0.489)/5 = 2.460/5 = **0.492 g/m**. CV = 0.0039/0.492 × 100 = **0.79%**. Limits = 0.492 ± 3 × 0.0039 = **0.480 to 0.504 g/m**.
2. UCL = 20.0 + 3 × 0.3 = 20.9. The reading 21.1 exceeds the UCL, so yes — treat it as an assignable cause and investigate the process, not the sample.
3. Readings inside the limits are random noise around a stable mean. Adjusting the process for each one adds the adjustment's own error to the natural spread (over-control or "tampering"), so total variation grows. Correct action is to leave a controlled process alone and act only on out-of-limit signals or clear trends.

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